SAWSAN ABDULBARI

Junior Data Scientist & Machine Learning Engineer | AI Innovation

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ABOUT

Data Scientist & ML Engineer with **3+ years of proven expertise** in building AI solutions that have positive business impact and can be scaled with business growth. **Worked in a global team of 15+ contributors** and **collaborated in globally distributed teams**, delivering **production-ready ML models** that solve challenges in healthcare, sustainability, and technology sectors. **Trilingual professional** (Arabic, English, Finnish) with **entrepreneurial background** and **passion for climate change solutions**.

Key Differentiators: Global collaboration (distributed teams, 15+ contributor projects), Production Expertise (deployed ML models), Business Knowledge (7-year entrepreneurial experience), Climate Focus (environmental sustainability advocate), Academic Excellence (student mentor & curriculum developer).

CORE COMPETENCIES

Machine Learning

- Predictive Modeling
- Time Series Analysis
- RAG Architecture
- Model Deployment
- Prompt Engineering

Data Science

- Data Mining & Analytics
- EDA & Visualization
- Feature Engineering
- Statistical Analysis
- Dashboard Creation

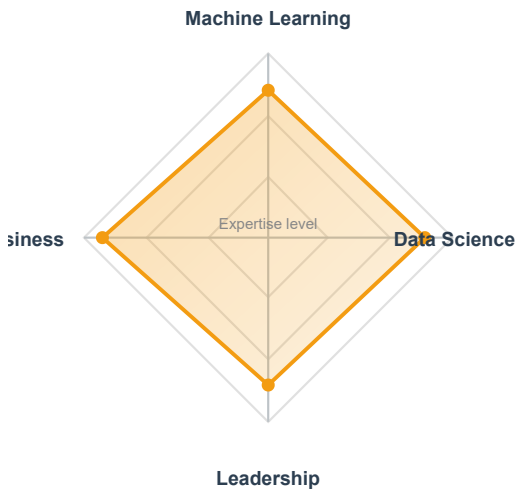
Leadership

- Team collaboration (15+ people)
- Cross-cultural Collaboration
- Mentoring & Training
- Project Management
- Stakeholder Communication

Business

- Strategic Planning
- Data Analysis
- Process Optimization
- Financial Management
- Business Development
- Operations Management

EXPERTISE MAPPING



Machine Learning	80%	**Data Science**	85%
Leadership	80%	**Business**	90%

Relative scale aligned with portfolio skills visualization (self-assessment).

PROFESSIONAL EXPERIENCE

Machine Learning Engineer
2024 - Present

Omdena | Palo Alto, CA (Remote)

Collaborating on AI innovation in globally distributed teams. Researching and implementing RAG architecture for enterprise applications.

Python | RAG Architecture | Machine Learning | AI Research

Student Assistant
2024 - 2025

Häme University of Applied Sciences (HAMK) | Finland

Improved educational outcomes for 50+ students and developed curriculum. Provided personalized guidance and technical mentoring.

Python | Teaching | Curriculum Development

Lead Machine Learning Engineer
2024

Omdena Kitwe, Zambia | Remote

Delivered a COVID-19 forecasting system for the Kitwe, Zambia (Omdena) chapter: XGBoost models, Streamlit app, and standard regression metrics (MAE, RMSE, R^2) as in the public repository. Worked with a global team of 15+ contributors.

Team Leadership | XGBoost | Healthcare AI

Machine Learning Engineer
2023 - 2024

Raqameyyah Digital Consultancy | Egypt (Remote)

Built and deployed ML models using advanced techniques. Developed chatbots with advanced prompt engineering and created front-end applications to enhance customer experience.

Machine Learning | Feature Engineering | Application Deployment | Python

Data Analyst Trainee
2023

Raqameyyah Digital Consultancy | Egypt (Remote)

Cleaned and preprocessed large datasets for analysis. Developed applications for data filtering and visualization and created automated data processing pipelines.

Data Analysis | Dashboards | Streamlit | EDA

Accounting Intern
2022

foodora Finland | Tampere

Gained experience in financial accounting and supply operations. Worked with Microsoft Excel for data management and participated in financial reporting processes.

Supply Operations | Financial Accounting | Microsoft Excel

Social Media Representative
2022

Foody Allen | Tampere

Acted as main content creator for food producer Njunno during Block festival weekend. Focused primarily on creating photography and video content from the event for marketing and post-marketing purposes.

Multi-Channel Marketing | Graphic Design | Event Photography | Video Editing

Marketing Communications and Content Production
2022

BSB Fashion | Paphos, Cyprus

Developed marketing strategies and content production. Managed e-commerce platforms and multi-channel marketing and created graphic designs and video content.

Sales | E-Commerce | Multi-Channel Marketing | Graphic Design

Sales Associate
2021 - 2022

Keinuipuiston kukka Oy | Tampere, Finland

Main responsibility was customer service and sales. Managed the store's e-commerce and delivery platforms Foodora and Wolt and developed business-to-business relationships.

Business-to-Business (B2B) | Sales | E-Commerce | Multi-channel Retail

Founder
2007 - 2014

Plastic Recycling for Sustainable Solutions | Hybrid

Founded and managed a plastic recycling factory focused on environmental protection. Developed business strategies and operational methods and managed employees and production processes. Promoted environmental protection by transforming plastic waste into valuable resources.

Business-to-Business (B2B) | New Business Development | Supply Operations | Environmental Sustainability

EDUCATION

Bachelor of Business Information Technology (Data Science) — Grade 4.7

Häme University of Applied Sciences (HAMK), Finland

2023 - 2026

Specialization in Data Science and Business Intelligence. Focus on machine learning, data analytics, and business applications.

Thesis: BITCA (bachelor thesis; PDF available for download on the portfolio website).

Climate Change AI (CCAI) Summer School

Climate Change AI (CCAI)

2024

Intensive program on machine learning and AI for climate change mitigation, adaptation, and environmental decision-making. Certificate: *Climate_Change_AI_Sawsan_Abdulbari.pdf* on the portfolio site.

Google Data Analytics Specialization

Google

Nov 2023

Courses: Foundations: Data, Data, Everywhere; Ask Questions to Make Data-Driven Decisions; Prepare Data for Exploration; Process Data from Dirty to Clean; Analyze Data to Answer Questions; Share Data Through the Art of Visualization; Google Data Analytics Capstone: Complete a Case Study. Certificate PDF: *Google_Data_Analytics.pdf* on the portfolio site.

Vocational Qualification in Business (Business Administration)

Tampere Vocational College (Tredu), Finland

2020 - 2022

Business administration and management fundamentals. Project management and organizational skills.

KEY PROJECTS

📊 Data Diwan — trade decisions, not trade signals

Vite, React, TypeScript, Supabase, Tailwind, Stripe, i18next, Vitest, Playwright

Trading app: position sizing, risk and discipline tools, plans and strategies, trade history, Stripe, PWA, automated tests and monitoring. https://datadiwan.com/

🏦 Kiva Loan EDA Dashboard

Python, Pandas, Plotly, Streamlit

Interactive dashboard analyzing Kiva microloan data with comprehensive EDA features. Includes loan distribution analysis, country-wise insights, and borrower demographics.

🦠 COVID-19 Case Prediction System

Python, XGBoost, scikit-learn, Streamlit, pandas

COVID-19 imputed cases and deaths for the Kitwe, Zambia (Omdena) chapter: trained XGBoost models, multi-page Streamlit app, and MAE/RMSE/R^2 evaluation as documented on GitHub.

🏠 California Housing Price Predictor

Python, Scikit-learn, Pandas, Streamlit

Machine learning model predicting California housing prices using regression algorithms. Features data preprocessing, feature engineering, and model comparison.

👥 Customer Segmentation Dashboard

Python, Scikit-learn, Plotly, Streamlit

Advanced customer segmentation using K-means clustering and RFM analysis. Interactive dashboard for business insights with actionable recommendations.

🚕 NYC Taxi Trip Duration Prediction

Python, XGBoost, Folium, Streamlit

Predicts taxi trip duration in NYC using machine learning algorithms. Incorporates geospatial features, weather data, and traffic patterns.

💵 Bank Note Authentication

Python, scikit-learn, Streamlit, pandas

UCI banknote (four wavelet features): compared SVM, k-NN, logistic regression, and decision trees, with cross-validation and a Streamlit deployment per the public repository.

🛒 E-commerce Analytics Dashboard

Python, Pandas, Plotly, Streamlit

Comprehensive e-commerce analytics dashboard providing insights into sales performance, customer behavior, and product trends with interactive visualizations.

🤖 Smart AI Document Chatbot with RAG

Python, FAISS, sentence-transformers, PyTorch, Gradio

RAG with FAISS retrieval, sentence embeddings, and an open-LLM path (e.g. Gemma-2B) documented in the public README; notebooks and a Hugging Face Space demo. Optional API keys for local experiments.

📰 News Sentiment & Keyword Analysis Dashboard

Python, NLTK, Pandas, Streamlit

Comprehensive dashboard for analyzing news sentiment and extracting key insights from articles using NLTK VADER sentiment analysis and advanced NLP techniques.

😊 Emotion Recognition Project

Python, PyTorch, EfficientNet, OpenCV

Advanced emotion recognition system using EfficientNet-B0 with comprehensive model optimization pipeline. Features multi-phase development including model selection on FER-2013 dataset, fine-tuning on RAF-DB dataset, and dynamic INT8 quantization for mobile deployment.

🌱 Carbon Fraud Detection Network

Python, FastAPI, Google Cloud, Vertex AI, PostgreSQL, Redis

Advanced AI-powered system for detecting carbon fraud using personality-based consensus agents and multi-source data integration. Features real-time fraud detection, adversarial AI agents, and comprehensive emission validation through CLIMATIQ API, OpenWeather, and satellite data integration.

🍳 AI Recipe Finder

Python, Whisper, Groq, Gradio

Voice-powered recipe app: say ingredients aloud and get AI-generated recipes. Uses Whisper for speech-to-text and Groq (or local FLAN-T5) for generation, built with Gradio and deployed on Hugging Face Spaces.

🏥 PIRHA Pre-Procedure Rehabilitation Guide

Vercel, Web, Healthcare UX

Web guide for pre-procedure rehabilitation for the Pirkanmaa wellbeing region (PIRHA). Supports patients and families before medical procedures. Live at ict-project-pirha.vercel.app.

🌳 Tree Classifier

Python, Machine Learning, Vercel, Web, Mobile

Interactive web and mobile app for tree species classification using machine learning. Live demo on Vercel.

LANGUAGES

Arabic	English	Finnish
Native	Professional proficiency	Professional proficiency

PERSONAL INTERESTS

- 🥾 **Hiking** - Nature exploration, outdoor activities
- 💪 **Fitness** - Health and wellness, physical activity
- 🍽️ **Food Exploration** - Culinary experiences, cultural cuisine
- 🎓 **Continuous Learning** - Technology trends, professional development
- 🌱 **Environmental Protection** - Green initiatives, environmentally friendly practices